

Class Activity: Designing a Logical Database

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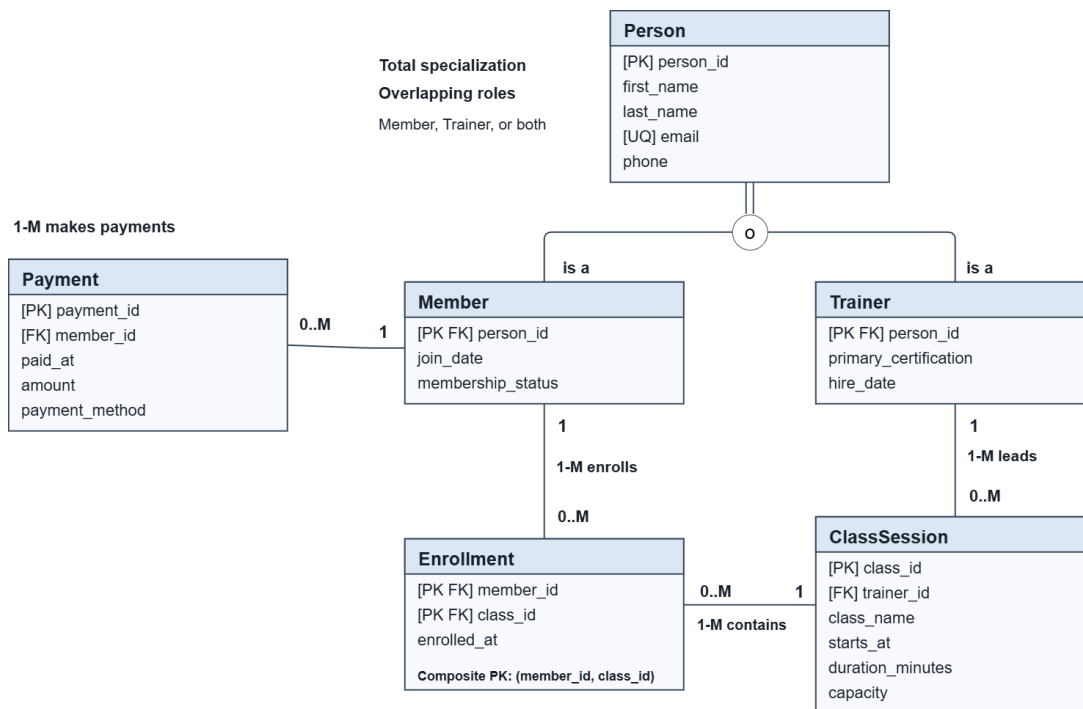
Entities:

- Supertype:
 - o Person
- Subtypes:
 - o Trainer
 - o Member
- Other Entities:
 - o Payment
 - o ClassSession
 - o Enrollment

Data Interaction Rules:

1. Every Person must be a member, trainer or both
2. A Member can make 0 or many payments, while each Payment belongs to one Member
3. A Member can enroll in zero or many ClassSessions
4. A ClassSession can have zero or many Members enrolled
5. A Trainer can lead zero or many ClassSessions, while each ClassSession is led by only one trainer
6. Member and Trainer share the Person entity's person_id as their primary and foreign key

EERD Diagram



PARTICIPATION AND KEY

M = many; every M side allows zero (0..M).
 Each subtype role is optional individually (0..1).
 Every Person is a Member, a Trainer, or both.
 Each subtype shares its Person key (1-1 key mapping).
 PK = primary key; FK = foreign key; UQ = unique.
 o = overlapping; double line = total specialization.

Data Dictionary

Person

Attribute	Definition	Data Type	Validation Rules
person_id	Unique id for a person	Integer	Primary key, required, unique
first_name	First name of a person	VARCHAR(50)	Required
last_name	Last name of a person	VARCHAR(50)	Required
email	Email address of a person	VARCHAR(100)	Required, unique, in valid email format
phone	Phone number of a person	VARCHAR(20)	Optional, in valid phone number format of region

Member

Attribute	Definition	Data Type	Validation Rules
person_id	Unique id for a member	Integer	Primary key, foreign key, references Person.person_id, required
join_date	Date a person became a member	DATE	Required, valid date format
membership_status	Current membership status of a member	VARCHAR(20)	Required

Trainer

Attribute	Definition	Data Type	Validation Rules
person_id	Unique id for a trainer	Integer	Primary key, foreign key, references Person.person_id, required
primary_certification	Main certification held by a trainer	VARCHAR(100)	Required
hire_date	Date trainer was hired	DATE	Required, valid date format

Payment

Attribute	Definition	Data Type	Validation Rules
payment_id	Unique id for a payment	Integer	Primary key, required, unique
member_id	Identifies which member made the payment	Integer	Foreign key referencing the member's person_id, required
paid_at	Date and time the payment was made	TIMESTAMP	Required, valid date and time format
amount	Amount of money paid	DECIMAL(10,2)	Required, greater than 0
payment_method	Method of payment	VARCHAR(30)	Required, valid payment method

ClassSession

Attribute	Definition	Data Type	Validation Rules
class_id	Unique id for each class	Integer	Primary key, required, unique
trainer_id	Identifies which trainer is leading the class	Integer	Foreign key references Trainer.person_id, required
class_name	Name of the class	VARCHAR(100)	Required
starts_at	Date and time class begins	TIMESTAMP	Required, valid date and time format
duration_minutes	Length of the class in minutes	Integer	Required, greater than 0
capacity	Max number of members allowed in the class	Integer	Required, greater than 0

Enrollment

Attribute	Definition	Data Type	Validation Rules
member_id	Identifies which member is enrolled in the class	Integer	Primary key, foreign key referencing Member.person_id
class_id	Identifies which class session a member is enrolled in	Integer	Primary key, foreign key referencing ClassSession.class_id

enrolled_at	Date and time a member was enrolled in a class	TIMESTAMP	Required, valid date and time format
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Business Rules

Business Rule 1: People and roles

- Each Person has one unique person_id and one unique email address. Every stored Person must be a Member, a Trainer, or both. This is a total, overlapping specialization: one individual can hold both roles while sharing the same name and contact details.

Business Rule 2: Membership

- Each Member belongs to exactly one Person and uses that Person's person_id as both its primary key and foreign key. A membership has a join date and a status of active or inactive. Only active members may create new enrollments. Historical enrollment and payment records are retained when a membership becomes inactive.

Business Rule 3: Scheduled classes

- Each ClassSession represents one scheduled occurrence with a class name, start date and time, positive duration and positive capacity. Exactly one Trainer leads each session. A Trainer may lead zero or many sessions. Repeated occurrences of the same class name have different class_id values.

Business Rule 4: Class enrollment

- A Member may enroll in zero or many ClassSessions, and each ClassSession may have zero or many members. Enrollment resolves this M-M relationship. Each Enrollment belongs to exactly one Member and one ClassSession. The composite primary key (member_id, class_id) prevents duplicate enrollment in the same session. The number of enrollments must not exceed the session capacity.

Business Rule 5: Completed payments

- Each Payment records one completed, positive payment from exactly one Member. A Member may have zero or many payments. All amounts are in CAD, and payment_method must be cash, debit or credit. Payments are independent of class enrollment; a payment does not create or identify a class booking.

Business Rule 6: Trainer details

- Each Trainer belongs to exactly one Person and uses that Person's person_id as both its primary key and foreign key. The model records one primary certification and one hire date per Trainer. Trainer-specific details are stored separately from shared personal details and membership details.

Relationships and Participation

Relationship	Type	Participation
Person to Member	ISA; 1-1 shared-key mapping	1 Person per Member; 0..1 Member role per Person
Person to Trainer	ISA; 1-1 shared-key mapping	1 Person per Trainer; 0..1 Trainer role per Person
Trainer to ClassSession	1-M	1 Trainer per session; 0..M sessions per Trainer
Member to Enrollment	1-M	1 Member per enrollment; 0..M enrollments per Member
ClassSession to Enrollment	1-M	1 session per enrollment; 0..M enrollments per session
Member to Payment	1-M	1 Member per payment; 0..M payments per Member
Member to ClassSession	M-M at the business level	0..M on both sides; implemented through Enrollment

Normalization and Design Rationale

- **1NF:**
 - o Each field holds only one value. Class selections are separate Enrollment rows rather than a list inside Member. Only one primary certification is recorded per Trainer.
- **2NF:**
 - o Enrollment uses the whole pair (member_id, class_id) as its primary key. Its non-key attribute, enrolled_at, describes that specific enrollment and depends on both IDs. All other entities have single-attribute primary keys.
- **3NF:**
 - o Shared contact details only appear in Person. ClassSession stores the trainer_id rather than only trainer details, and Payment stores the member_id rather than only member details. No non-key attribute determines another non-key attribute.
- **Scope:**
 - o class_name is a session label, not a unique catalogue key. It doesn't determine the duration, capacity of the class or the trainer. The payment amount is the recorded transaction amount, not a price determined by the payment_method.

Team contribution

Team Member	Role/Contribution
Troy Franks	Set up GitHub repo Business rules
Seth Garciano	EERD and proofread
Tristin Bates	Entity listing, Data Interaction Rules, Data Dictionary and Document Formatting
Yisong Wang	Business Rules, EERD based on the business rules, Relationships and Participation and Normalization and Design Rationale